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Evaluation and quality control of personal nephelometers in indoor, outdoor and personal environments

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Personal nephelometers provide useful real-time measurements of airborne particulate matter (PM). Recent studies have applied this tool to assess personal exposures and related health effects. However, a thorough quality control (QC) procedure for data collected from such a device in a large-scale exposure assessment study is lacking. We have evaluated the performance of a personal nephelometer (personal DataRAM or pDR) in the field. We present here a series of *post hoc* QC procedures for improving the quality of the pDR data. The correlations and the ratios between the pDRs and the collocated gravimetric measurements were used as indices of the pDR data quality. The pDR was operated in four modes: passive (no pump), active (with personal sampling pumps), active with a heated inlet, and a humidistat. The pDRs were worn by 21 asthmatic children, placed at their residences indoors and outdoors, as well as at a central site. All fixed-site pDRs were collocated with Harvard Impactors for $PM_{2.5}$ ($HI_{2.5}$). By examining the differences between the time-weighted average concentrations calculated from the real-time pDRs' readings and recorded internally by the pDRs, we identified 9.1% of the pDRs' measurements suffered from negative drifts. By comparing the pDRs' daily base level with the $HI_{2.5}$ measurements, we identified 5.7% of the pDRs' measurements suffered from positive drifts. High relative humidity (RH) affected outdoor pDR measurements, even when a heater was used. Results from a series of chamber experiments suggest that the heated air stream cooled significantly after leaving the heater and entering the pDR light-scattering chamber. An RH correction equation was applied to the pDR measurements to remove the RH effect. The final R^2 values between the fixed-site pDRs and the collocated $HI_{2.5}$ measurements ranged between 0.53 and 0.72. We concluded that with a carefully developed QC procedure, personal nephelometers can provide high-quality data for assessing PM exposures on subjects and at fixed locations. We also recommend that outdoor pDRs be operated in the active mode without a heater and that the RH effect be corrected with an RH correction equation.
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Introduction

To monitor short-term variation and identify sources of personal exposure to particulate matter (PM), a real-time instrument is needed. The personal DataRAM (pDR-1000, Thermo Andersen, Smyrna, GA, USA) is one of the commercially available personal nephelometers, and has been used widely in many recent and ongoing exposure and health effect studies (Muraleedharan and Radojevic, 2000; Quintana et al., 2000; Williams et al., 2000; Quintana et al., 2001; Rea et al., 2001; Liu et al., 2002; Wallace et al., 2003). Although previous studies have validated the pDR as a useful personal PM monitor, several drawbacks remain. The response of the light-scattering devices depends on the size, density, shape, and optical properties of the sampled aerosols. Such devices need to be calibrated against gravimetric devices in the field, rather than relying on the calibration curve developed by the manufacturer with standard test dust. In addition, light-scattering is affected by relative humidity (RH) (McMurry and Stolzenburg, 1989; Thomas and Gebhart, 1994; Brauer, 1995; Day et al., 2000; Quintana et al., 2000; Day and Malm, 2001). Previous papers showed that the RH effect may be corrected by either physical (Sloane, 1984; Sloane and Wolff, 1985; Lowenthal et al., 1995; Sioutas et al., 2000) or statistical models (Richards et al., 1999), or prevented by using a heater (Liu et al., 2002) or a diffusion drier (Sioutas et al., 2000). Another issue pertaining to pDR is the baseline or zero drift, which appears to result from particle contamination of the light source chamber. Although Liu et al. (2002) reported minimal zero drift ($<2\,\mu g/m^3$), Howard-Reed et al. (2000)

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reported a daily baseline drift from nine pDRs worn by elderly subjects for two 24-h periods ranging from −3 to 6 $\mu g/m^3$. Quintana et al. (2001) reported a baseline drift of approximately 40 $\mu g/m^3$ for one pDR carried for seven consecutive days.

In this study, the pDR was used to monitor PM exposures among young children and at their residences in Alpine, CA, USA. Although we had tested the pDR extensively (Quintana et al., 2000, 2001) before deployment in the field, we found unexpected outliers and baseline drifts in the final field measurements. In light of these pDR measurement errors, we developed a series of physically justifiable strategies to identify and remedy these problems, and evaluated the pDR performance after each quality control (QC) strategy.

Methods

Field Study Design and Instruments

This study was conducted in fall 1999, consisting of two sampling sessions, and spring 2000, consisting of five sampling sessions. Each sampling session lasted for 14 consecutive days. During each session, PM concentrations (recorded at 1-min intervals) were measured using the pDRs on up to four subjects, inside and outside at their residences, and at a central site. An RH and temperature sensor (HOBO® data logger, Onset Computer Corp., Pocasset, MA, USA) was attached on top of each pDR, also recording at 1-min intervals. At all fixed-locations, the pDRs were collocated with 24-h integrated Harvard Impactors for $PM_{2.5}$ ($HI_{2.5}$) (Air Diagnostics and Engineering, Inc., Naples, ME, USA) using 37 mm Teflon filters (2.0 μm PTFE membrane, Gelman Laboratories, Ann Arbor, MI, USA) operated at a flow rate of 10 l/min. The $HI_{2.5}$ data were used as the reference for the pDR measurements. Both Babich et al. (2000) and Liu et al. (2003) reported a good agreement between the $HI_{2.5}$ and the federal reference method for $PM_{2.5}$ ($R^2 > 0.97$). Although the pDRs were not preceded by a $PM_{2.5}$ inlet, Quintana et al. (2000) and Liu et al. (2002) reported high correlation between the pDR and HI $PM_{2.5}$ ($R^2 = 0.66 \sim 0.84$).

In all, 21 asthmatic children wore the pDRs in a fanny pack at waist height during waking hours, and placed the instrument on a table near their breathing zone while asleep. The indoor pDRs were located where the participants spent most of their time, such as the living room, bedroom, or other main activity areas. The home outdoor pDRs were located under an open rain shield in the yard at least 1 m away from the walls. The monitor inlets were approximately 0.9 m above the ground at home sites.

The pDRs were operated in four modes (Quintana et al., 2000): (1) *passive (PAS)*: This is the default configuration of the pDR; (2) *active and unheated (AUH)*: A personal sampling pump (Model 224, SKC Inc., Eighty Four, PA, USA) drew 2 l/min of air through the pDR's sampling chamber to avoid potential undersampling in stagnant air; (3) *active and heated (AH)*: A heater (DR-TCH, Thermo Andersen, Smyrna, GA, USA) was attached to the pDR inlet and it raised the ambient temperature by 30°F to avoid RH artefacts. This temperature increment reduced the RH to below 50%, as estimated from a standard psychrometric chart (MIE, 1994); and (4) *AH with a humidistat that switches on and off the electricity based on a pre-set RH level (AHH)*: This mode is the same as the AH mode except that the heater was set to be activated only when the ambient RH was over 85%. Although several studies (e.g., Thomas and Gebhart, 1994; Sioutas et al., 2000) observed the RH effect starting at 60%, our previous field tests (Quintana et al., 2000) showed that the RH effect on the pDR in the AUH mode was moderate between 60% and 85% and prominent above 85%. This, in conjunction with the concern of losing particle-bound semivolatile compounds due to heating the incoming air all the time (Bergin et al., 1997), prompted us to set the humidistat at 85%. For personal and home indoor monitoring, the pDRs were operated in the PAS mode. For home outdoor and central site monitoring, the AUH and AH modes were used in the two fall sessions, whereas the AHH mode was used in all five spring sessions. An additional pDR in the PAS mode was located at the central site during both the fall and spring sessions.

The sampling procedures in the fall sessions were described in detail in Quintana et al. (2000). Briefly, the pDRs and $HI_{2.5}$ were operated over 24-h periods from early evening to early evening. The pDR was zeroed at the beginning of each runday with the zeroing bag (Z-Pouch) attached to a small HEPA filter as supplied by the manufacturer. Data were downloaded at the end of each runday. All filters were weighed on a microbalance (Model 30, Cahn Instruments, Madison, WI, USA) and corrected for field blanks (one indoor blank and one outdoor blank per runday). A similar sampling procedure was used in the spring sessions except for the zeroing frequency. In considering that the manufacturer's zeroing set may not provide adequate zero air, the pDR was zeroed only at the beginning of each session with zero air through a HEPA filter connected to a pump (air flow rate: 4 l/min). Subjects were trained to use an electronic diary programmed on Palm® personal digital assistants (Palm, Inc., Milpitas, CA, USA) to report their times, locations, and activities (time-activity diary, or TAD) with a 15-min resolution.

The 1-min pDR measurements ($pDR_{1\,min}$) were averaged over 15-min intervals ($pDR_{15\,min}$) to match the TAD information. The sample sizes for the $pDR_{15\,min}$ on the subjects, at home indoors, at home outdoors, and at the central site were 25,930, 26,420, 26,505, and 9067, respectively. They were also averaged over 24 hours ($pDR_{24\,h}$) with on and off times corresponding to the collocated $HI_{2.5}$ sampling period. The SAS statistical program (Release 8, SAS Institute, Inc., Cary, NC, USA) was used for data analysis.

Chamber Experiments

From the QC procedures (as described in the next section), it was identified that there might be some operational issues associated with the heaters and humidistats. To test the performance of the heaters, a series of validation experiments was conducted inside a 46 cm (W) × 76 cm (L) × 51 cm (H) chamber, with a commercially available humidifier (Model # DH-903, Duracraft Corp., Whitinsville, MA, USA) adjusting RH between 58% and 95%. Two sampling trains were set up. One sampling train included a pump drawing the air through the heater continuously at 2 l/min, with a NIST traceable thermohygrometer (Model # 03313-66, Cole-Parmer, Vernon Hills, IL, USA) measuring the temperature and RH at the outlet of the heater. The second sampling train included a pump drawing the air through the heater and the pDR continuously at 2 l/min. The thermohygrometer measured the temperature and RH at the outlet of the pDR. Two pDRs were used in the second sampling train. A leak test indicated that the two pDRs had an air leak of 0.6% and 20.4%, respectively, at a 2 l/min flow rate. During the experiments, a HOBO® data logger with built-in temperature and RH sensors recorded the temperature and RH at the heater's air inlet. Furthermore, three humidistats were tested in the same chamber at RH between 90% and 95% with the dials set at 85%. Light bulbs were connected to the electrical outlets on the humidistats to determine if the line voltage were activated.

QC Procedures

We developed a series of *post hoc* procedures to identify pDR artefacts and drifts and strategies to resolve these measurement problems.

Identification of Outliers

A simple time-series plot and a univariate analysis of the $pDR_{1\,min}$ data helped identify potential outliers, especially unexplainable and unjustifiable extreme values. In one case, $pDR_{1\,min}$ increased from 39 to 7390 $\mu g/m^3$ in 1 min and then decreased to 133 $\mu g/m^3$ the next minute. As the TAD data showed no potential PM generating activities, such a phenomenon was very likely due to pDR artefacts. In light of these extreme values, we developed two screening criteria to identify them. The first criterion identifies physically impossible values, that is, the $pDR_{1\,min}$ remained over 4000 $\mu g/m^3$ (>99th percentile) for more than 2 min. The second criterion identifies unlikely values based on natural exponential decays, that is, the $pDR_{1\,min}$ was greater than 2000 $\mu g/m^3$ and at least 25 times (>99th percentile) greater than the previous or subsequent $pDR_{1\,min}$. All identified outliers were confirmed by checking the TAD data to ensure that no PM-generating events or activities were reported. Since these identified outliers may still contain partial information about the actual PM levels, they were substituted with smoothed filter values derived from a low-pass linear filter (Shumway, 1988) based on a 31-min center-weighted moving average, in which the weighting factor for the outliers was 23.7%. Although only less than 0.01% of the $pDR_{1\,min}$ outliers were detected, they affected the $pDR_{15\,min}$ values significantly ($P<0.01$ for paired t-test between the $pDR_{15\,min}$ calculated with or without $pDR_{1\,min}$ outliers). In the example given earlier, the $pDR_{15\,min}$ for that time period was 572.6 $\mu g/m^3$ when the outlier (i.e., 7390 $\mu g/m^3$) was included and was 85.6 and 198.1 $\mu g/m^3$ when the outlier was deleted or replaced with the smoothed value (i.e., 1687.5 $\mu g/m^3$), respectively.

Even after smoothing the $pDR_{1\,min}$ outliers, potential outliers were still apparent in the $pDR_{15\,min}$ time-series data. We used two similar but more stringent criteria applied to the smoothed 15-min averages to identify extreme $pDR_{15\,min}$ values that were not supported by any reported personal or indoor activities: (1) the $pDR_{15\,min}$ was greater than 2000 $\mu g/m^3$ and (2) the $pDR_{15\,min}$ was at least 10 times greater than the previous or subsequent $pDR_{15\,min}$. Both criteria represent values greater than 99th percentile of the distribution. Only seven $pDR_{15\,min}$ outliers (0.03%) were found and removed from the dataset.

Identify Negative Baseline Drift

The pDR comes with a built-in function that reports the time-weighted average concentration ($TWAC_{pDR}$) of PM during the monitor operation period. Such averages can also be calculated manually once the measurements have been downloaded ($TWAC_{man}$). Although $TWAC_{pDR}$ should be identical to $TWAC_{man}$, this is only the case when there is no baseline drift below 0 $\mu g/m^3$. Such a negative drift may be caused by zeroing errors (e.g., contamination of the zero air) or by change of the environmental conditions. The pDR's internal memory that determines the $TWAC_{pDR}$ registers negative signals, whereas the memory that stores the time-series measurements only records zero or positive values. For example, if the baseline drift is -5 $\mu g/m^3$ when the true PM concentration is 3 $\mu g/m^3$, the internal memory would register -2 $\mu g/m^3$ while the external memory logs 0 $\mu g/m^3$. As long as the value of the negative drift is greater than the ambient concentration, the 24-h $TWAC_{man}$ would be greater than the 24-h $TWAC_{pDR}$. In all, 31 observations (3.5%) were identified where the 24-h $TWAC_{pDR}$ was 2 $\mu g/m^3$ or more lower than the calculated 24-h $TWAC_{man}$ value (Table 1). Incidentally, these 24-h averages were from $pDR_{1\,min}$ time series registering more than 20% (or $\geq$5 h) of zero values. There were 11 (1.2%) additional zero 24-h $TWAC_{pDR}$ with a high frequency of zero $pDR_{1\,min}$ (>58%) during the rundays (Table 1). Such a high frequency of zero values measured by the pDRs in the field often implies a negative drift problem. However, the frequency of zero $pDR_{1\,min}$ should not be used as the sole evidence of negative drifts since

Table 1. Summary table for the 24-h average observations of (A) ($TWAC_{pDR} - TWAC_{man}$) $< -2\,\mu g/m^3$, and (B) $TWAC_{pDR} = 0$ but ($TWAC_{pDR} - TWAC_{man}$) $> -2\,\mu g/m^3$.

Location	$TWAC_{pDR}$[a] ($\mu g/m^3$)	$TWAC_{man}$[b] ($\mu g/m^3$)	DIFF[c] ($\mu g/m^3$)	$HI_{2.5}$ ($\mu g/m^3$)	Zero Freq(%)[d]
(A) Personal	0.0	3.8	−3.8	—	1085 (77)
	0.0	3.6	−3.6	—	1039 (71)
	2.0	6.7	−4.7	—	940 (69)
	2.0	6.3	−4.3	—	862 (60)
	16.0	19.5	−3.5	—	817 (54)
	2.0	6.1	−4.1	—	735 (52)
	6.0	15.0	−9.0	—	727 (54)
	24.0	26.5	−2.5	—	570 (41)
	2.0	6.7	−4.7	—	437 (32)
	18.0	20.6	−2.6	—	427 (30)
	8.0	10.9	−2.9	—	393 (31)
Indoor	0.0	3.6	−3.6	8.4	1056 (72)
	0.0	2.8	−2.8	7.4	889 (62)
	1.0	3.6	−2.6	5.4	841 (58)
	0.0	3.7	−3.7	11.1	808 (59)
	2.0	5.2	−3.2	9.2	803 (56)
	0.0	4.0	−4.0	10.3	738 (49)
	1.0	6.6	−5.6	11.5	724 (53)
	0.0	6.3	−6.3	9.4	703 (49)
	1.0	3.8	−2.8	4.9	690 (49)
	1.0	5.5	−4.5	10.0	610 (43)
	0.0	5.1	−5.1	10.9	588 (41)
	2.0	5.7	−3.7	12.1	437 (31)
	12.0	14.6	−2.6	13.0	430 (30)
	5.0	7.6	−2.6	13.0	296 (21)
Outdoor	0.0	4.5	−4.5	14.1	1000 (71)
	0.0	4.6	−4.6	10.3	840 (48)
	7.0	18.3	−11.3	15.6	775 (55)
	1.0	3.7	−2.7	1.7	666 (42)
	15.0	21.5	−6.5	20.1	468 (33)
	24.0	31.1	−7.1	24.1	415 (35)
(B) Indoor	0.0	1.3	−1.3	5.6	1040 (75)
	0.0	1.8	−1.8	7.5	942 (65)
Outdoor	0.0	0.2	−0.2	4.8	1310 (97)
	0.0	0.2	−0.2	4.8	1310 (97)
	0.0	0.6	−0.6	3.5	1000 (67)
	0.0	0.6	−0.6	3.5	1000 (67)
	0.0	1.2	−1.2	6.2	875 (58)
	0.0	0.3	−0.3	11.9	1324 (89)
	0.0	1.2	−1.2	4.2	1198 (88)
	0.0	0.6	−0.6	10.1	982 (69)
	0.0	1.0	−1.0	9.3	680 (63)

[a]Time-weighted average concentrations reported from the pDR's built-in function.
[b]Time-weighted average concentrations calculated from the $pDR_{1\,min}$.
[c]$TWAC_{pDR}$–$TWAC_{man}$.
[d]Frequency (and percentage) of zero $pDR_{1\,min}$ during one runday.

days of low ambient PM levels exist. In our data set, there were four observations (mean $HI_{2.5} = 3.8\,\mu g/m^3$) with more than 20% zero $pDR_{1\,min}$ during the rundays, while the $TWAC_{pDR}$ was neither zero nor lower than the $TWAC_{man}$. Currently, it is not possible to estimate the magnitude of such a negative bias and when it occurred, thus time series with evidence of negative drift were removed from analysis.

The identification of the negative drift was further evaluated by comparing the $pDR_{24\,h}$ with the collocated $HI_{2.5}$ measurements. The pDR responds most efficiently to particles with aerodynamic diameter of $0.6\,\mu m$, with the sensitivity falling off to about 16% at diameters larger than $5\,\mu m$ and to about 10% at diameters below $0.15\,\mu m$. Since the pDR also responds to particles with diameters larger than $2.5\,\mu m$, its measurements are expected to be greater than those of $HI_{2.5}$. Furthermore, assuming that aerosol density is the major different characteristic between the ambient and calibration aerosols, the density of the aerosols sampled by the pDRs can be estimated as 2.6*($HI_{2.5}$/pDR), where $2.6\,g/cm^3$ is the density of the calibration dusts used by the pDR manufacturer (Liu et al., 2002). Previous studies (Ames et al., 2000) estimated a density of $1.4\,g/cm^3$ for organic carbon (OC) and $2.0\,g/cm^3$ for elemental carbon (EC), and the density of ammonium nitrate and ammonium sulfate is 1.72 and $1.77\,g/cm^3$, respectively (O'Neil, 2001). As $PM_{2.5}$ in southern California (Downey, CA, USA) consisted of 6% EC, 46% OC, 24% ammonium nitrate and 16% ammonium sulfate (Sioutas et al., 2000), a reasonable estimate for $PM_{2.5}$ density weighted by composition is $1.5\,g/cm^3$ and possible values could range between 1.4 (100% OC) and $1.8\,g/cm^3$ (100% ammonium sulfate). Consequently, the average ratio of pDR/$HI_{2.5}$ should be approximately 1.7 or range between 1.4 and 1.9. This estimated range also should be applicable to most cities in the US since sulfate, nitrate and organic carbon are the major outdoor $PM_{2.5}$ components as measured by the IMPROVE network (EPA, 1999) and the PM Speciation Trend Network (EPA, 2003).

Although light-scattering responses depend on many factors other than particle density, previous studies have shown that the pDR is on average 1.5 times (which falls within our estimated range of 1.4–1.9) higher than the collocated gravimetric measurements across homes and cites (Howard-Reed et al., 2000; Liu et al., 2002; Wallace et al., 2003), or

$$pDR_{24\,h} = \sum_{i=1}^{96} pDR_{15\,min,\,i} \approx 1.5PM_{2.5} \qquad (1)$$

Having a $pDR_{24\,h}$ value smaller than the $HI_{2.5}$ indicates that either the density of the measured aerosols was greater than $2.6\,g/cm^3$, which is unlikely, or the pDR underestimated the concentration of the ambient PM due to a negative drift. In all, 24 (77%) fixed-site pDR observations that meet the previous criteria, that is, ($TWAC_{pDR}$–$TWAC_{man} \leq -2$) or ($TWAC_{pDR} = 0$) are also $2\,\mu g/m^3$ lower than the collocated $HI_{2.5}$ measurements (Table 1), providing confirmation of the negative drift identification criteria. In addition, 25 (4.1%) fixed-site $pDR_{24\,h}$ values were found to be $2\,\mu g/m^3$ lower

than the collocated $HI_{2.5}$ values when both measurements were well above the limit of detection of the pDR, that is, $2\,\mu g/m^3$ (Wallace et al., 2003). These 25 observations passed the ($TWAC_{pDR}$–$TWAC_{man} \leq -2$) criteria probably because the value of the negative drift was smaller than the ambient concentration ($HI_{2.5} = 6.6$–$23.5\,\mu g/m^3$).

Positive Baseline Drift

Positive baseline drifts occur when the pDR light-scattering chamber is contaminated. In this case, the pDR overestimates $PM_{2.5}$ levels, resulting in an unreasonably low estimated particle density. We identified the potential positive drifts by comparing the pDR's daily "base level" with the collocated 24-h $HI_{2.5}$ measurements. The pDR's daily base level was calculated based on a 75-min interval, defined as the minimum of 5-point centered moving average of $pDR_{15\,min}$ during each runday. The lowest $pDR_{15\,min}$ during any 24-h monitoring period, or the pDR's daily base level should be lower than 1.5 times the collocated 24-h $PM_{2.5}$ value for Eq. (1) to be valid. Taking into account the measurement uncertainty and allowing for fluctuations in the density of ambient aerosols, we defined a positive drift when the pDR's daily base level was $> 2.6 * HI_{2.5}$. The observations that do not pass this criterion will have the estimated particle density at an unreasonably low range ($< 1\,g/cm^3$). An example of positive drifts is shown in Figure 1, where the $pDR_{24\,h}$ is $170.7\,\mu g/m^3$, the pDR's daily base level was estimated as $70.6\,\mu g/m^3$, and the $HI_{2.5}$ was $19.6\,\mu g/m^3$. Currently, the absolute value of the positive drift could not be estimated because collocated real-time gravimetric measurements were not available in this study. We identified a positive drift for 35 (5.7%) fixed-site observations and removed them from analysis. For these observations, the average ratio of pDR's daily base level to $HI_{2.5}$ was 3.49, as compared to 0.88 for the rest of the observations.

For personal pDRs, which were not collocated with any gravimetric measurements, the positive drift was identified using the "collocated" home indoor pDRs when the subjects were asleep. Personal pDR measurements from six subjects were found to be consistently higher than the collocated indoor pDRs. Figure 2 shows an example of an elevated personal pDR as compared with the collocated indoor pDR. The magnitude of the positive drift or the correction factor for a personal pDR can be estimated using the average difference between the indoor and personal pDRs. Then the personal pDR's readings were subtracted by the correction factor to give the calibrated personal pDR measurements. As not all subjects have the indoor pDRs placed in the bedrooms (e.g., in the living rooms), occasional excursions occurred on one pDR but not on the other (e.g., when subjects stirred at night). Measurements during such excursions should not be included in the determination of the correction factor. Excursions were defined when (1) there is a change in one pDR but not in the other pDR. That is, the difference between the $pDR_{15\,min,\,t}$ at time t and $pDR_{15\,min,\,t-1}$ or $pDR_{15\,min,\,t+1}$ is greater than $5\,\mu g/m^3$ for one of the two collocated pDRs, but the difference among the $pDR_{15\,min,\,t}$, $pDR_{15\,min,\,t-1}$, $pDR_{15\,min,\,t+1}$ for the second pDR stayed within $2\,\mu g/m^3$; and (2) there are excessive fluctuations in personal or indoor pDRs. That is, the difference between $pDR_{15\,min,\,t}$ and $pDR_{15\,min,\,t-1}$ or $pDR_{15\,min,\,t+1}$ is greater than $20\,\mu g/m^3$ (> the 99th percentile). In Figure 2, the personal pDR15 min between 0300 and 0400 hours (filled symbols) were identified as excursions according to the first criterion. For the six pDRs, 7.4% (range: 3.7–11.9%) of the pDR15 min were identified as excursion data points. The correction factor ranged between 13.5 and $129.6\,\mu g/m^3$ ($\text{mean} \pm \text{SD} = 76.2 \pm 33.0\,\mu g/m^3$).

RH Effects and Heater Performance

Pearson's correlation coefficients between the RH and fixed-site $pDR_{15\,min}$ in the PAS (indoor), AUH (outdoor), AH (outdoor) and AHH (outdoor) modes are 0.20, 0.26, 0.32, 0.67 and 0.27, respectively. Controlling for the actual

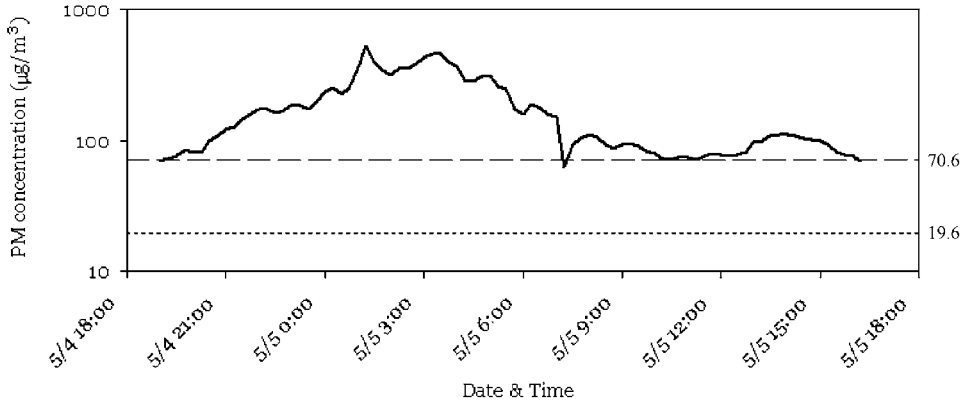


Figure 1. An example of the pDRs that experienced the positive baseline drifts. The solid line is the $pDR_{15\,min}$ data. The pDR daily base level (dashed line) was estimated as the minimum 75-min centered moving average of $pDR_{15\,min}$ during the 24-h period. The dotted line is the $HI_{2.5}$ measurement.

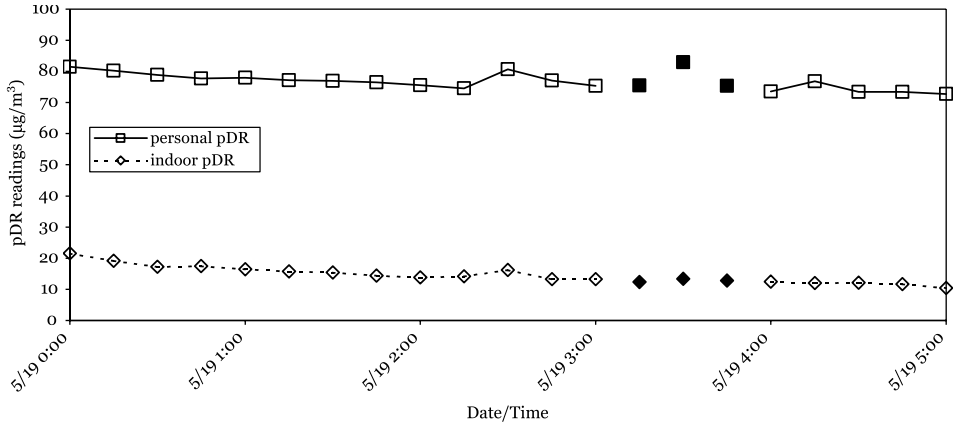


Figure 2. An example of the personal pDR that experienced the positive baseline drifts. The $pDR_{15\,min}$ between 0300 and 0400 hours were the identified excursions points (filled symbols) and were not included for calculating the correction factor of the personal pDRs.

ambient PM concentrations, significant positive correlations between the $pDR/HI_{2.5}$ ratio and RH were found for outdoor pDRs in the AUH (Figure 3b), AH (Figure 3c) and AHH (Figure 3d) modes ($P<0.05$). This indicates that the heating system did not effectively reduce the RH to the expected level. In addition, the mean ratio of the collocated 1-h pDR in the PAS mode and 1-h pDR in the AH mode at the central site at RH $>60\%$ was 1.11 ± 0.17 ($r=0.89$). This suggests that the RH effects persisted for pDRs in the AH mode and provides further evidences that the heater did not function properly. For indoor pDRs, the $pDR/HI_{2.5}$ ratio and RH was not correlated (Figure 3a) because 97% of the time the indoor RH was below 60% (mean = 36.6%, SD = 12.8%).

The RH effect could be corrected using physical models when the particle size and composition are known (Sloane, 1984; Sloane and Wolff, 1985; Lowenthal et al., 1995; Sioutas et al., 2000). Richards et al. (1999) also constructed a simple regression model to correct for the RH effects without knowing the aerosol size and composition:

$$\ln\left(\frac{PM_{2.5_Bsp}}{PM_{2.5_true}}\right) = -a\ln(1-RH) - b \qquad (2)$$

where $PM_{2.5_Bsp}$ is $PM_{2.5}$ measured by a light-scattering device; $PM_{2.5_true}$ is the RH corrected (true) $PM_{2.5}$ concentration; and a and b are empirically determined parameters. They used the measurements from a heated nephelometer as the $PM_{2.5_true}$ and the measurements from an unheated nephelometer as $PM_{2.5_Bsp}$. Two sets of parameters were calculated $\{(a,b) = (0.68,\ 0.35)$ and $(0.31,\ 0.01)\}$ based on the RH readings reported from an RH sensor located at the monitoring site and on the nephelometer, respectively. Figures 3e–h show the correlations between the $pDR/HI_{2.5}$ ratio and RH after correcting for the RH effect using the first set of parameters for $pDR_{15\,min}$ measured between 60% and 95% RH. Correction for pDR measurements taken at RH below 60% is not necessary as such RH level has little effect on the pDRs (Thomas and Gebhart, 1994; Quintana et al., 2000). At high RH ($>95\%$) the correction curve is too sensitive to RH such that an accurate correction is not possible (Richards et al., 1999). Thus $pDR_{15\,min}$ data taken at RH $>95\%$ were removed. After the RH correction, no correlation was observed between the $pDR/HI_{2.5}$ ratio and RH for outdoor pDRs in the AUH (Figure 3f) and AHH (Figure 3h) modes. Although the correlation remained significant for the AH mode, the R^2 decreased from 0.64 (Figure 3c) to 0.25 (Figure 3g). Note that using the second set of parameters by Richards et al. (1999), the corrected pDRs also showed decreased correlations between the $pDR/HI_{2.5}$ ratio and RH ($R^2 = 0.04$, 0.41, 0.04 for outdoor pDRs in the AUH, AH, and AHH modes, respectively).

Results and discussion

Table 2 summarizes the comparison results between the collocated fixed site $pDR_{24\,h}$ and $HI_{2.5}$ after each data QC procedure. N indicates the number of remaining samples. We used the coefficient of determination, R^2, between the $pDR_{24\,h}$ and $HI_{2.5}$ and median of the $pDR_{24\,h}/HI_{2.5}$ ratio to determine the adequacy of each QC procedure. The R^2 generally increased after each QC procedure. Figure 4 shows the correlations between the $pDR_{24\,h}$ and $HI_{2.5}$ after these QC procedures. Both Spearman's and Pearson's correlation coefficients were calculated for $pDR_{24\,h}$ and $HI_{2.5}$. The differences between the two correlation coefficients are all smaller than 0.03. The final median $pDR/HI_{2.5}$ ratios for all pDR types were between 1.2 and 1.9 (Table 2). For the indoor pDRs in the PAS mode, the R^2 and median $pDR/HI_{2.5}$ ratio did not change much after the RH correction. This is mainly because the indoor RH was usually below 60%. For the AUH mode, the RH correction increased the

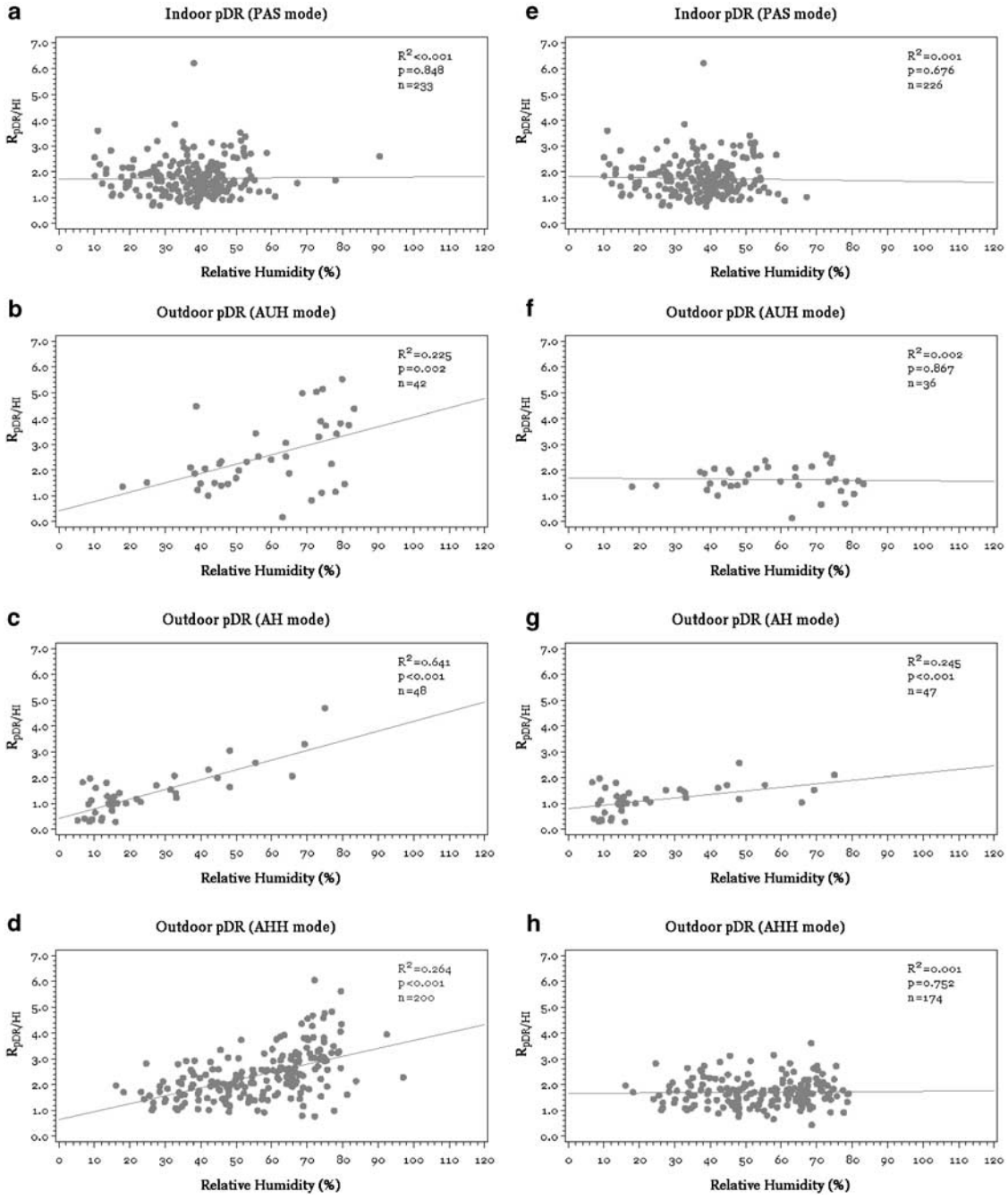


Figure 3. Scatter plots of the pDR/HI ratio vs. RH before (a–d) and after (e–f) the RH corrections using Eq. (1) with the first set of parameters developed by Richards et al. (1999) (PAS: passive; AUH: active unheated; AH: active with a heated inlet; AHH: active with a heated inlet and a humidistat).

R^2 from 0.4 to 0.7, and the median $pDR/HI_{2.5}$ ratio decreased from 2.3 to 1.6.

For the pDRs in the AH mode, the median $pDR/HI_{2.5}$ ratio was 1.2 (Table 2, Column V.1), suggesting that the $PM_{2.5}$ levels were underestimated even after all the QC procedures. This could be due to the removal of semivolatile compounds by overheating from the heaters (Bergin et al., 1997). At RH < 60%, collocated pDRs running in the PAS and AH modes at the central site were expected to have comparable readings. However, the mean ratio of the 1-h pDR measurements in the PAS mode to the AH mode was 1.34 (SD = 0.45) at RH < 60%. Figure 5 shows that high ratios generally occurred at high ambient temperature (> 80°F) and low RH (< 15%). The ratio (mean = 1.55,

Table 2. Comparisons between the $pDR_{24\,h}$ and $HI_{2.5}$ after each data quality control and correction procedure.

		Cumulative QC procedures[c]					
pDR mode[a]	Variable	I	II	III	IV	V.1	V.2
PAS (indoor)	*N*	260	244	234	233	226	226
	Median of $pDR_{24\,h}/HI_{2.5}$	1.6	1.6	1.7	1.7	1.7	1.7
	R^{2b}	0.50	0.50	0.53	0.55	0.54	0.54
AUH (outdoor)	*N*	48	48	45	42	36	36
	Median of $pDR_{24\,h}/HI_{2.5}$	2.3	2.3	2.4	2.3	1.6	1.9
	R^{2b}	0.35	0.35	0.40	0.41	0.72	0.73
AH (outdoor)	*N*	60	55	49	48	47 (20)[d]	47 (20)[d]
	Median of $pDR_{24\,h}/HI_{2.5}$	1.1	1.1	1.2	1.2	1.2 (1.5)	1.2 (1.5)
	R^{2b}	0.50	0.48	0.53	0.60	0.70 (0.53)	0.68 (0.48)
AHH (outdoor)	*N*	246	236	230	200	174	174
	Median of $pDR_{24\,h}/HI_{2.5}$	2.3	2.4	2.4	2.3	1.7	1.8
	R^{2b}	0.31	0.33	0.36	0.53	0.62	0.62

[a]PAS: passive; AUH: active unheated; AH: active with a heated inlet; AHH: active with a heated inlet and a humidistat.
[b]R^2 values between $pDR_{24\,h}$ and $HI_{2.5}$.
[c]I: Correction for outliers.
II: I & correction for negative baseline drifts (compared to $TWAC_{pDR}$).
III: II & correction for negative baseline drifts (compared to $HI_{2.5}$).
IV: III & correction for positive baseline drifts.
V.1: IV & RH correction with the 1st set of parameters in Richards *et al*. (1999).
V.2: IV & RH correction with the 2nd set of parameters in Richards *et al*. (1999).
[d]Numbers in parentheses are statistics after removing $pDR_{15\,min}$ with temperature $>80^{\circ}F$ and RH $<15\%$.

SD = 0.65) was significantly higher at ambient temperature $>80^{\circ}F$ and RH $<15\%$ than that (mean = 1.24, SD = 0.26) at the other range of temperature and RH. Removing these points resulted in a reasonable $pDR/HI_{2.5}$ ratio of 1.5 (Table 2, Column V.1). However, the mass loss due to semivolatile components of the aerosol, that is, the ''overheating effect'', is a function of temperature, RH, and particle composition. Thus, using the above criterion alone may result in unnecessary removal of good data points, as evidenced by the lower R^2 in Table 2 (Column V.1, compared to the R^2 before correcting for the overheating effect). The correlation between the $pDR/HI_{2.5}$ ratio and RH in Figure 3g became insignificant ($R^2 = 0.02$, $P = 0.58$) after correcting for the overheating effect.

In correcting for the RH effect (Eq. (2)), we applied the ambient RH measured by the HOBO® attached to the top of the pDR, assuming that the heating system on the pDR was not effective. Our chamber experiments showed an effective reduction of 32% in RH between the inlet and outlet of the heater. However, the RH reduction between the heater's inlet and the pDR's outlet is only 10% and 8% for the pDR with 1% and 20% air leakage, respectively. Evidently, air cooled substantially after entering the pDR, effectively raising the RH inside the pDR. This resulted partially in the observed RH effect on pDRs in the AH mode (Figure 3c). The air leakage of the active pDRs also accounted for the RH effect. Therefore, the ambient RH is a good surrogate for the RH inside the pDR. Note that the actual RH inside the pDR could not be measured — our modeling attempt based on an exponential decay curve from our limited experimental data failed to produce a better R^2 than those using the ambient RH. Compared to the air cooling phenomenon inside the pDRs, the degree of RH effect due to the air leakage may be less substantial, judging from the small difference in the pDRs' RH reduction (8% vs. 10%) between the 1% and 20% of air leakage in our chamber experiments.

For pDRs in the AHH mode, the RH correction increased the R^2 between $pDR_{24\,h}$ and $HI_{2.5}$ by 0.1 and decreased the median $pDR/HI_{2.5}$ ratio from 2.3 to 1.7 (Table 2, Column V.1). Although the humidistat was set to activate the heater at ambient RH $>85\%$, our chamber experiments indicated that only one out of three humidistats activated the line voltage when RH $>$ 90%. These humidistats were probably not suitable or designed for applications in high RH conditions. In addition, the R^2 and median $pDR/HI_{2.5}$ ratio in the AHH mode were more similar to those in the AUH mode than those in the AH mode (Table 2). It is thus likely that most humidistats did not activate the heaters at high RH. Therefore, the ambient RH was a good surrogate for the RH inside the pDR when applying the RH correction equation (Eq. (2)). Note that

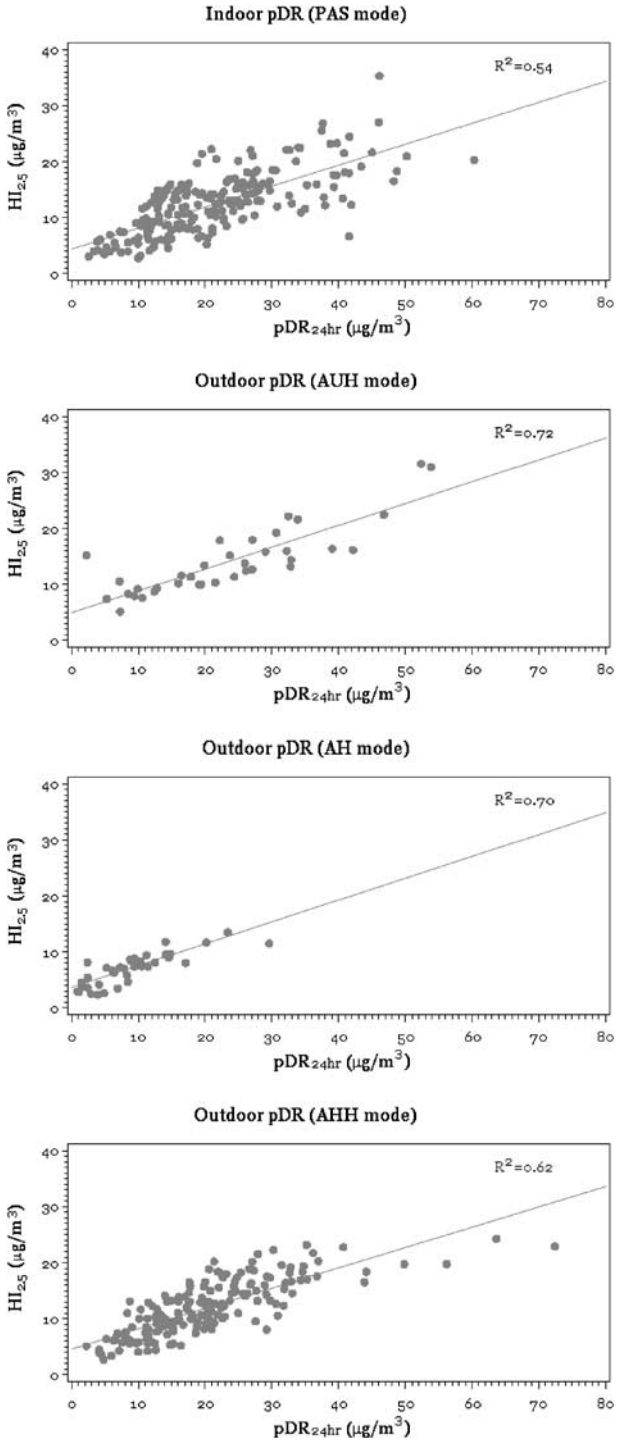


Figure 4. Scatter plots of the pDR_{24h} vs. $HI_{2.5}$ after the RH corrections using Eq. (1) with the first set of parameters developed by Richards et al. (1999) (PAS: passive; AUH: active unheated; AH: active with a heated inlet; AHH: active with a heated inlet and a humidistat).

since the heaters were off most of the time, pDRs in the AHH mode were not expected to be influenced by the overheating effect.

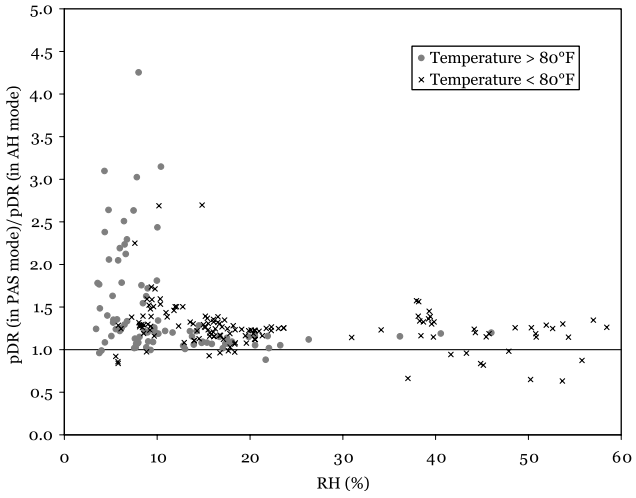


Figure 5. Ratio of 1-h pDRs (in the PAS mode) to 1-h pDRs (in the AH mode) as a function of RH at the central site at RH < 60%.

Although collocated pDRs in the PAS and AHH modes at the central site were expected to have comparable readings at RH < 60%, 10% of the observations had PAS/AHH ratios higher than 2. A weak but significantly positive correlation (Pearson's correlation coefficient = 0.13, $P = 0.002$, $n = 588$) was found between the ratio and wind speed, suggesting that the passive pDR may over sample under high wind conditions. Nevertheless, since such a wind effect is infrequent, our reported overheating effect for the pDRs in the AH mode at low RH still holds.

The active pDR is a sealed system. The air leakage found in our study could be attributed to loose connectors or damage of the sealant near the optical chamber assembly. Loose connectors can be easily fixed by the field technicians, but problems involving internal components need to be repaired by the manufacturer. It is essential to check the air leak on a regular basis to minimize the RH effects from the air leakage.

The choice of the screening criteria for outlier identification should depend on the distribution of the $pDR_{1\,min}$ data and the characteristics of the study subjects and homes. In this present study of nonsmokers living in nonsmoking residences, 4000 $\mu g/m^3$ was used because it was above the 99th percentile of the $pDR_{1\,min}$ measurements. This screening criterion for outliers was further supported by results of Howard-Reed et al. (2000) and Quintana et al. (2001), who reported that the maximum $pDR_{1\,min}$ measurements were approximately 2900 $\mu g/m^3$. This screening criterion should be shifted upward for study populations or regions with prominent PM sources. Wallace et al. (2003) reported in their Inner City Asthma Study that the 1-h pDR measurements were over 4000 $\mu g/m^3$ in the smoking residences. Balakrishnan et al. (2002) also reported that the personal exposures (average sampling duration = 2.7 h) to respirable

PM for cooks using wood fuels during cooking times in rural India could be higher than 4000 $\mu g/m^3$.

Both sets of parameters reported by Richards et al. (1999) were applied in Eq. (2). The first set reduced the RH effect more successfully than the second set. Although these two sets of parameters were calculated from data collected in Bakersfield, CA, reasonable $pDR/HI_{2.5}$ ratios and good R^2 values between $pDR_{24\,h}$ and $HI_{2.5}$ after the RH corrections (Table 2) indicated that they were applicable in this study. Previous studies have also reported that the chemical composition of $PM_{2.5}$ did not affect significantly the light-scattering response of the DataRAM (Sioutas et al., 2000).

Light-scattering responses depend on particle density, refractive index, and particles size (Sioutas et al., 2000). Since the refractive index is similar across major outdoor $PM_{2.5}$ species (1.47–1.55) (Sioutas et al., 2000) and the calibration dust (1.5), the accuracy of the particle density estimation equation (i.e., $2.6*HI_{2.5}/pDR$) is affected by the particle size difference between the calibration dust (mass media diameter: 2–3 μm) and the measured PM.

In our QC procedures, we identified the baseline drifts before correcting for the RH effects as drifts occurred throughout the sampling period while the RH artefacts most likely occurred during the nighttime. Performing RH correction before the baseline drift identification decreased the final R^2 values for $pDR_{24\,h}$ and $HI_{2.5}$ from 0.72 to 0.63 (AUH mode) and 0.62 to 0.57 (AHH mode), while the R^2 values for the PAS and AH modes were approximately the same.

To identify the negative drifts, we first compared the $TWAC_{pDR}$ to $TWAC_{man}$. This approach may not provide sufficient sensitivity to identify all the observations with negative drifts, especially those with small drifts. There may be cases when the negative drift produced negative numbers, but there were not enough of them to create a 2 $\mu g/m^3$ difference during a 24-h period. To examine the extent of this issue, we simulated 520 24-h time series at 1-min resolution with a geometric mean ranging between 5 and 30 $\mu g/m^3$ with a 1 $\mu g/m^3$ interval, and a geometric standard deviation ranging between 1.1 and 3.0 with a 0.1 interval. The spline-smoothed lines of the $TWAC_{man}$–$TWAC_{pDR}$ values and percent of zero $pDR_{1\,min}$ of the simulated time series with different drift values were presented in Figures 6a and b, respectively. The percent of zero $pDR_{1\,min}$ required to produce a 2 $\mu g/m^3$ difference in $TWAC_{man}$–$TWAC_{pDR}$ depends on both the true $PM_{2.5}$ and the drift values. The simulation results indicated that the ($TWAC_{pDR}$–$TWAC_{man} > -2$) criterion underestimated the percentage of pDR's having negative drifts at high ambient $PM_{2.5}$ levels and at small negative drift values ($< 8\ \mu g/m^3$). At true $PM_{2.5}$ of 10 $\mu g/m^3$, this criterion identifies observations with negative drift values greater than 8 $\mu g/m^3$ (Figure 6a), where there are approximately 52% of zero $pDR_{1\,min}$ (Figure 6b). When the true $PM_{2.5}$ is 20 $\mu g/m^3$, this criterion only identifies most observations with negative drift values greater than 12 $\mu g/m^3$ (Figure 6a), where there are approximately 38% of zero $pDR_{1\,min}$ (Figure 6b).

The large positive drifts found for the personal pDRs were attributed to elevated personal activities and infrequent zeroing. Based on the activity levels recorded on the TAD (i.e. levels 1–5 for activity levels ranging from sleep, rest, light, moderate to strenuous), four of the six subjects carrying pDRs with positive drifts were more active (mean level = 2.0–2.2) than the other 17 subjects (mean level = 1.3–1.9). This also explains why the positive drifts observed in our study were higher than those reported earlier (Howard-Reed et al., 2000; Liu et al., 2002). Howard-Reed et al. (2000) reported that the highest baseline drift was 6 $\mu g/m^3$, while Liu et al. (2002) observed minimal zero drift ($< 2\ \mu g/m^3$). This is because they had mostly elderly subjects who were less active than the child subjects and daily zeroing was deployed in both studies. Furthermore, all personal pDRs with positive drifts were operated in the spring sessions, during which the pDRs were zeroed only once every 14 days as compared to the daily zeroing in the fall sessions. This is also evident from the outdoor pDR measurements. In the fall sessions, the percentage of measurements with positive drifts was 6% and 1.7% for the pDRs in the AUH and AH modes, respectively. In the spring sessions, the rate of positive drifts was 12% for the pDRs in the AHH modes. This suggests that daily zeroing may alleviate positive drifts. The fact that the outdoor pDRs in the fall sessions were still affected by the positive drifts despite the daily zeroing procedures indicated that using the manufacturer-provided bag might not eliminate the drift problems totally. An improved zeroing protocol, such as using clean air with active sampling system, should be considered. If the pDRs are zeroed correctly at the beginning of each sampling day, the occurrence and magnitude of the baseline drift could be estimated by checking the results of the postexposure zeroing, and by comparing the ambient PM levels recorded by the pDRs after the 24-h sampling period to the readings from a newly zeroed pDR. In addition to the daily zeroing procedures, annual or semi-annual cleaning and recalibration of the pDRs by the manufacture could also help to remove the build-up (e.g. grease from cooking sources) inside the sensing chamber, thus alleviating serious zeroing problems.

Conclusion

The personal nephelometer is a valuable tool for accessing real-time PM exposures. Many operational issues of the pDR need to be identified and handled properly before data analysis. In this paper, we presented a series of QC procedures to identify pDR measurement artefacts. By comparing the $TWAC_{man}$ to $TWAC_{pDR}$ and $pDR_{24\,h}$ to

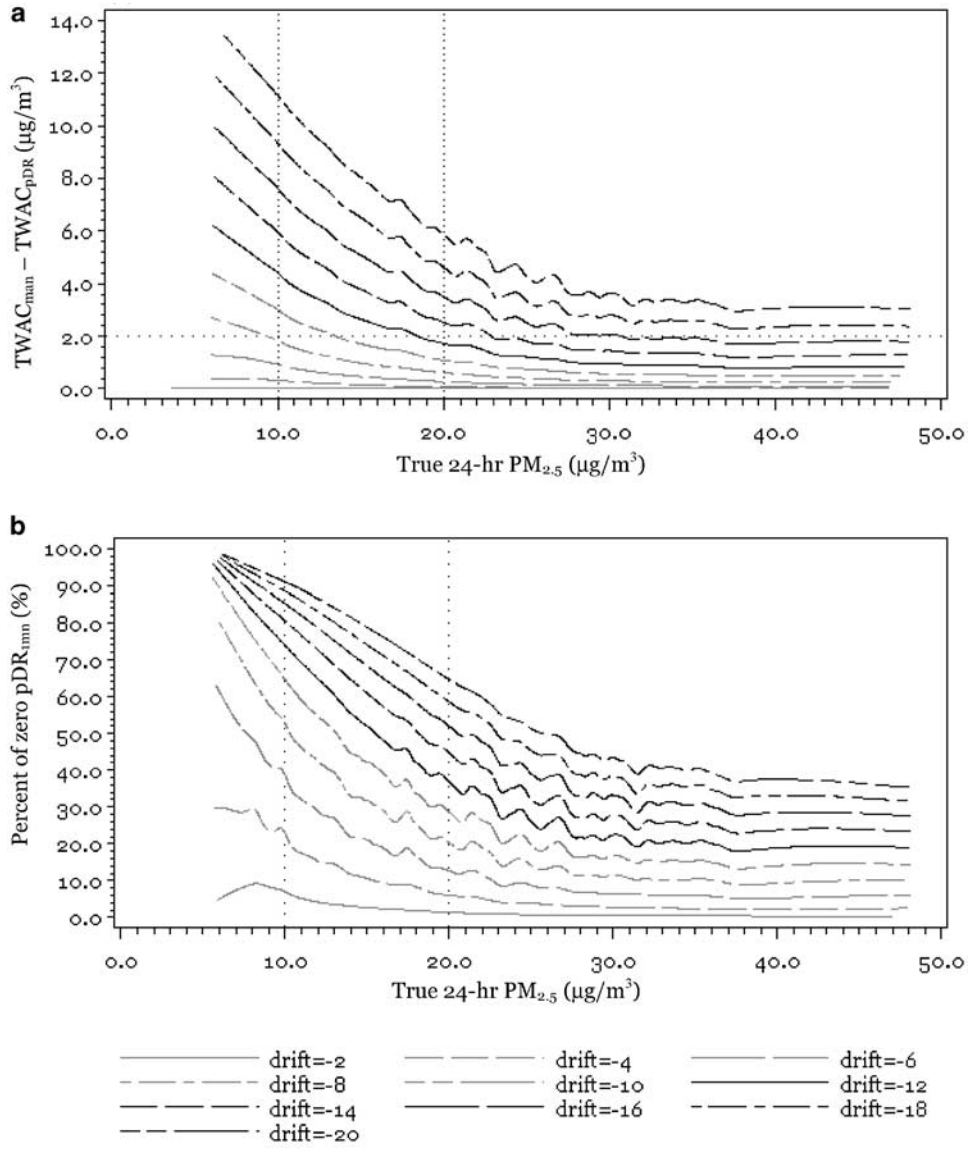


Figure 6. The spline-smoothed lines of (a) the $TWAC_{man}$–$TWAC_{pDR}$ values and (b) percent of zero $pDR_{1\,min}$ of the simulated time series with different drift values.

$HI_{2.5}$, negative drifts were identified. By comparing the pDR's daily base level to $HI_{2.5}$ measurements, positive drifts were identified. We successfully removed most RH effects by applying a statistical RH correction equation. The final results were satisfactory. Our R^2 for $HI_{2.5}$ and $pDR_{24\,h}$ are similar to the R^2 of 0.66 (personal filter samples vs. pDR) reported by Howard-Reed et al. (2000) but lower than the R^2 of 0.77–0.84 found by Liu et al. (2002). Our $pDR/HI_{2.5}$ ratios are similar to the ratios of 1.5–1.6 reported in other studies (Howard-Reed et al., 2000; Liu et al., 2002; Wallace et al., 2003). For identifying drifts of pDRs, we suggest to have collocated gravimetric measurements or consider using active pDRs with an integrated cyclone and filter (pDR-1200, MIE, Bedford, MA, USA). The performance of pDR-1200 can be found in Lanki et al. (2002). Although we attempted to minimize the potential zeroing errors in the field by zeroing the pDRs only at the beginning of each session with "true zero air" in the spring, our study results indicate that the benefit of daily zeroing of the pDR outweighs the zeroing errors.

The pDR in the passive mode is adequate for indoor PM monitoring, given the reasonable $pDR/HI_{2.5}$ ratio and R^2 between $pDR_{24\,h}$ and $HI_{2.5}$. The pDR in the passive mode is adequate for personal PM monitoring with our QC procedures and calibration against a fixed-site pDR or other gravimetric samplers. When applied outdoors, pDRs can be cost-effectively operated in the AUH mode. In the AH mode, our heater had difficulty effectively eliminating sufficient RH at high RH levels and overheated the air under hot and dry conditions. The RH effect for pDRs in the

AUH mode can be corrected with the RH correction equation. To entirely avoid the RH and overheating effects, an upstream diffusion drier (Sioutas et al., 2000) may be a good alternative, although such a device typically cannot operate for 24 h without replacement of the absorbing material (Quintana et al., 2000; Lanki et al., 2002).

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